12	**(a)**		because it's neither a one-to-one nor a many-to-one (mapping) do not allow eg because it's neither a one-to-one nor a many-to-one **function**	**B1**	**2.4**	allow because it's one-to-many (mapping) allow eg because each value of x is mapped to two values **oe** do not allow eg because it's a one-to-many **function**
				[1]		
12	**(b)**		$\frac{x^2+2}{x^2+2-3}$ $\frac{x^2+2}{x^2-1}$ or $\frac{x^2+2}{(x-1)(x+1)}$	**M1** **A1**	**1.1** **1.1**	
				[2]		

Question			Answer	Marks	AO	Guidance
12	(c)		$\|x\| > 1$ **or** $x < -1$ or $x > 1$ **or** $x < -1$, $x > 1$ **or** $x < -1 \cup x > 1$	**B1**	**1.1**	do not allow eg $x < -1$ and $x > 1$ eg $-1 > x > 1$
				[1]		